CO1-AT2 – Short Answers

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Course Code: ITA0329

Course: Mobile Computing

Questions:

1. Given

Bandwidth per full-duplex channel = 2*25=50 kHZ

Total bandwidth = 45*1000 = 45000 kHz

Number of channels = 45000 % 50 = 900

full-duplex channels = 900 channels

2. Given

Channels per cell = Total channels / Number of cells

= 560 / 7 = 80

Channels = 80 channels per cell.

3. Given

Total voice channels = Number of carriers × Time slots per carrier

= 100 × 8 = 800

800 simultaneous voice channels.

4. Given

Total channels = 42 × 1000 / 300 = 140

Channels per cell = 140 / 7 = 20

Result: 20 channels per cell.

5. Given

C/I = Carrier power - Interference power

= -60 - (-92) = 32 dB

Result: 32 dB.

6. Given

Total channels = Number of cells × Channels per cell

= 4 × 15 = 60

Result: 60 channels.

7. Given

Slot duration = Frame duration / Number of slots

= 5 / 8 = 0.625 ms

Result: 0.625 ms.

8. Given

Number of carriers = Total bandwidth / Carrier spacing

= 24 × 1000 / 200 = 120

Result: 120 carrier frequencies.

9. Given

Total capacity = Number of cells × Calls per cell

= 6 × 250 = 1500

Result: 1500 simultaneous calls.

10. Given

Average call duration = 3 × 60 = 180 seconds

Active calls = Call arrival rate × Average call duration

= 4000 × 180 = 720000

Result: 720000 active calls.

11. Given

Total bandwidth = Number of channels × Bandwidth per channel

= 150 × 25 = 3750 kHz

= 3.75 MHz

Result: 3.75 MHz.

12. Given

Number of guard bands = 30 - 1 = 29

Total guard-band bandwidth = 29 × 40

= 1160 kHz = 1.16 MHz

Result: 1.16 MHz.

13. Given

Cell diameter = 2 × 1.5 = 3 km

Handoffs per hour = Distance travelled / Cell diameter

= 90 / 3 = 30

Result: 30 handoffs per hour.

14. Given

Total bandwidth = 6 × 1000 = 6000 kHz

Number of channels = 6000 / 30 = 200

Result: 200 channels.

15. Given

Total bandwidth = 40 × 1000 = 40000 kHz

Number of users = 40000 / 400 = 100

Result: 100 users.

16. Given

Offered traffic = Call attempts per minute × Average call duration in minutes

= 40000 × 3 = 120000 Erlangs

Capacity = 100000 connections

Excess traffic = 120000 - 100000 = 20000 Erlangs

Excess fraction = (20000 / 120000) × 100

= 16.67%

Result: 16.67%

17. Given

Utilization = (Current load / Maximum capacity) × 100

= (9000 / 12000) × 100

= 75%

Result: 75% utilization.

18. Given

Total channels = 48 × 1000 / 200 = 240

Channels per cell = 240 / 4 = 60

Result: 60 channels per cell.

19. Given

Number of carriers = 30 × 1000 / 200

= 150

Result: 150 carrier frequencies.

20. Given

Path loss = Transmitted power - Received power

= 43 - (-87)

= 130 dB

Result: 130 dB.

21. Given

Total channels = 48 × 1000 / 200 = 240

Channels per cell = 240 / 6 = 40

Result: 40 channels per cell.

22. Given

Paging cycles = Total subscribers / Users per cycle

= 150000 / 6000 = 25

Result: 25 paging cycles.

23. Given

SNR = Signal power - Noise power

= -85 - (-115)

= 30 dB

Result: 30 dB.

24. Given

Total bandwidth = 250 × 12

= 3000 kHz

= 3 MHz

Result: 3 MHz.

25. Given

Total throughput = 1500 × 600

= 900000 kbps

= 900 Mbps

Result: 900 Mbps.